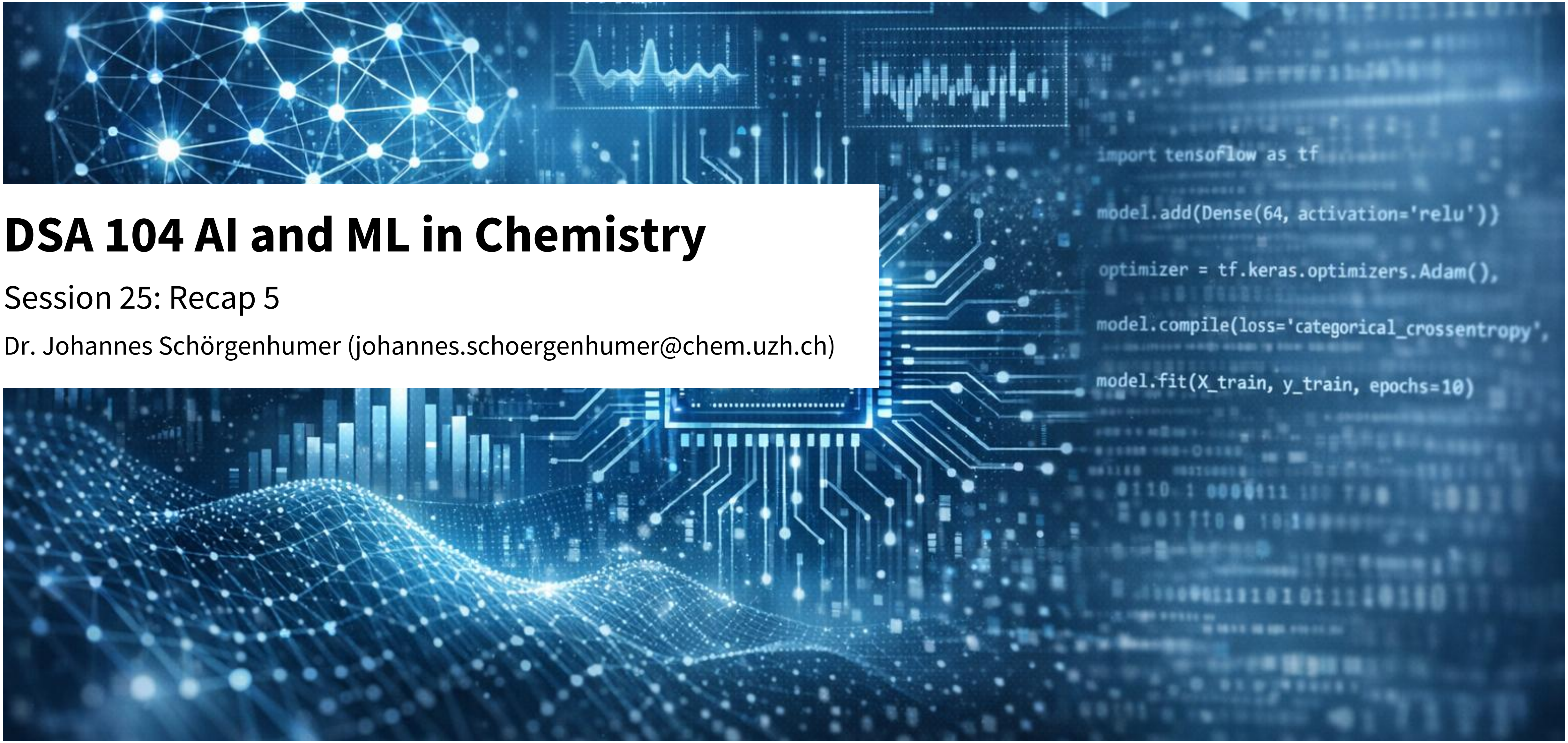
Universität Zürich UZH

Department of Chemistry



DSA 104 AI and ML in Chemistry

Session 25: Recap 5

Dr. Johannes Schörgenhumer (johannes.schoergenhumer@chem.uzh.ch)

Points for today

Discussion Exercise 10

Discussion Mock exam

Symposium

Outlook

Feedback and course discussion

Universität Zürich[UZH]

Department of Chemistry

DSA Symposium: Automation and Data Science in Action

Lecture room Y11-F-06

27 May 2026, 08:30 – 16:00

DSA Symposium – Agenda:

27 May 2026
Y11-F-06

08:30 – 09:15 *Advancing hair and nail matrix analysis with automated workflow solutions*
Tina Binz (UZH)

09:15 – 10:00 *Biocatalysis as a Tool in Synthesis: Continuous-Flow Routes to Greener Chemistry*
Glenn Bojanov (University of Bern)

10:00 – 10:30 *Coffee break*

10:30 – 11:45 *Tracking plant volatiles in real time: New tools for studying biotic interactions*
Tristan Cofer (University of Bern)

11:45 – 12:00 *Optimising Chemical Process Development with Automation and Machine Learning: An Industrial Perspective*
Joshua W. Sin (F. Hoffmann-La Roche Ltd)

12:00 – 13:00 *Lunch for speakers and registered participants*

13:00 – 13:40 *Machine learning for predictive ecotoxicology*
Christoph Schür (Eawag)

13:40 – 14:20 *Generative machine learning for catalyst discovery* (virtual lecture)
Sarina Kopf (EPFL)

14:20 – 15:00 *From automation to autonomy: the rise of Self-Driving Labs®*
Loïc Roch (Atinary Technologies Inc.)

15:00 – 16:00 *Apéro (Y38-G-32) and lab tours*

Taste for more? Get involved!

Basic knowledge of all the tools for your own data science…

— Now: Ready to deploy the knowledge! (Thesis!)

— Ready for a new challenge? **SDL Hackathon!!** (See last slide)

More courses of the DSA Minor – of course not only for chemists:

— DSA 102: Lab Automation and Accelerated Chemistry

— DSA 103: Advanced Chemical Data Science (4 ECTS, fall semester)

— DSA 105: Automation and Chemical Data Science in practice (4 ECTS)

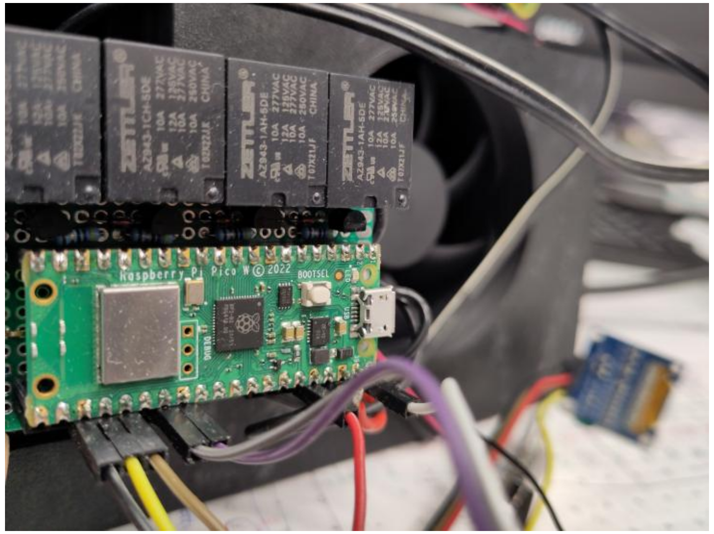
Exciting Electives:

— **NEW: DSA 301 Automating Chemistry – HTEL Live! (4 ECTS, DSA 102 needed)**

— CHE 725 Hacking for Chemists: Prototyping and Hardware Programming (2 ECTS, fall semester)

— **NEW:** DSA 302 Hacking the HTEL – Prototyping in Action! (2 ECTS, CHE 725 highly recommended)

Looking for a thesis? Join HTEL and be part of the revolution!



Why?

— Lab meets data science – high demand!!

— State-of-the-art infrastructure

— High-value collaborations with AI developers and digital chemistry groups

— Multidisciplinary research

— 3D prototyping and custom solutions

— **We have robots ☺**

What?

— Printing catalysts: Metal-ionic liquids

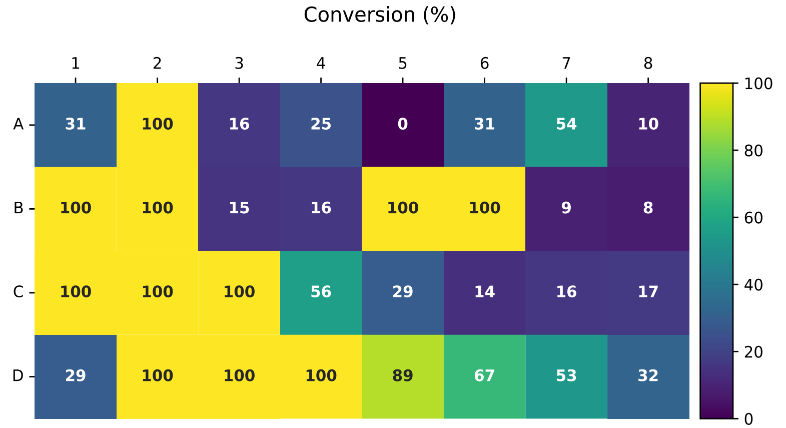
— Catalytic LEGO: Building-block catalysis

— Enriching big reaction data for AI agent development, e.g. Suzuki couplings

— Grow beautiful crystals: Development of HT crystallisation

— Getting the analysis down: Efficient HTE quantification

— Technical projects: low level robotics, custom integrations (OT2), ML applications, …



Self-driving labs Hackathon by EPFL LIAC and UZH HTEL

Interested in a new challenge?

Join the world's first practical lab automation hackathon!

How to participate:

1) Have a look at the announcement
2) Gather some teammates
3) Think of some cool research projects
4) Apply (same link as above) – deadline May 31
5) Just do it ☺
6) Win a prize?

Feedback and course discussion

Feedback?

Course format?

Guest contributions?

Take aways?